CONSTITUTIONAL FORCING

KHAYYAM WAKIL

ABSTRACT. We introduce and formally define *Constitutional Forcing*: a proposed mechanism by which the algebraic structure of a mathematical system would determine a governing constant, leaving no freedom in its value.

A system of depth k is defined to have exactly k independent *constitutional constraints*, operating on a binary configuration space of size 2^k, and we postulate that exactly k configuration classes are eliminated, leaving $2^k - k$ valid configurations. The governing constant is then the configuration density

$$\theta_k = \frac{2^k - k}{2^k}.$$

This count is a postulate, not a theorem: if a configuration is called invalid whenever some constraint fails, there are $2^k - 1$ invalid configurations, not k.

We examine three candidate instances. The Bombieri–Vinogradov level of distribution $1/2$ is assigned depth $k = 1$, but the assignment rests on the binary predicate "prime or not prime", and the same reasoning would assign $1/2$ to any predicate. The cascade moduli $q = 3^K$ are proposed as a depth-3 system with conjectured level $\theta_3 = 5/8$ [16]; this assignment is currently unsupported. Shannon's theorems [11] give the constants $H/\log|\mathcal{A}|$ and C, not $1/2$, and are retained only as a proposed analogy. We state the Constitutional Forcing Conjecture: every system with k independent constitutional constraints has $\theta_k = (2^k - k)/2^k$ as its governing constant, and every depth $k \geq 1$ is realised.

This paper is the first of a nine-paper programme (p01–p09). The remaining eight papers develop the proposed instances, their consequences and the cross-domain comparisons.

Revision note (November 2026). The Counting Theorem of the February version is withdrawn as a theorem: its induction miscounts (under the paper's own semantics there are $2^k - 1$ invalid configurations, not k), and the count k is now stated as a postulate (Postulate 2.6). The Kolmogorov "$\alpha = 3/4$" instance is withdrawn: the formula gives $\theta_2 = 1/2$, and Kolmogorov's $3/4$ is the exponent in $\eta/L \sim \mathrm{Re}^{-3/4}$, obtained by dimensional analysis [6]. The Bombieri–Vinogradov statement is corrected: it is an average over moduli, GRH does not raise its level, and levels above $1/2$ are known for well-factorable weights. The cascade-moduli instance is downgraded to a proposal: $\theta_W = 5/8$ is a conjecture, and the three "levels" modulo 3, 9 and 27 are not independent. The Shannon instance is downgraded to a proposed analogy, the Sophie Germain "non-instance" is reduced to a remark, and the Pascadi citation is corrected. The programme table now lists the nine papers p01–p09 with their revised status.

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1. What Constitutional Forcing Is

1.1. **The problem this paper addresses.** Some constants in mathematics invite the question "why this value and not another?" The Bombieri–Vinogradov exponent 1/2 is an example. This paper proposes a framework in which such a constant would be read off from the number of independent structural constraints on the system, and it states the framework precisely enough that each proposed instance can be checked.

We call it *Constitutional Forcing*. The name is deliberate. A constitutional constraint is one that is not imposed from outside the system but is *inherent to its constitution*—to what the system fundamentally is. Constitutional Forcing, if it occurs, is the situation in which a system's own constitution leaves no room for any governing constant other than the one it has.

1.2. **Status of the proposal.** The two arithmetic candidates examined in Section 3 rest, so far, on numerical coincidences: the Bombieri–Vinogradov exponent 1/2 equals θ_1, and the exponent 5/8 that Pascadi [9] obtained for primes with triply-well-factorable weights to general moduli equals θ_3. A coincidence of this kind is not evidence. Section 3 checks each candidate against the definition and records that none currently satisfies it in full.

The purpose of this paper is therefore to fix the definitions on which the eight companion papers rely, and to state precisely what would have to be proved for a system to count as an instance.

1.3. **Relationship to the Shannon–Wakil Effect.** In the programme's terminology, Constitutional Forcing is the proposed *mechanism* and the Shannon–Wakil Effect [18] the proposed *observable*: the reduction of an exponential configuration space to its valid part. The intended relationship is:

$$\text{Constitutional Forcing} \xrightarrow{\text{acts on}} \text{exponential space} \xrightarrow{\text{produces}} \text{Shannon–Wakil Effect.}$$

The present paper supplies the definitions on which [18] relies.

2. The Definitions

2.1. Constitutional constraints.

Definition 2.1 (Constitutional constraint)**.** Let S be a mathematical system with a binary configuration space at each operational level. A constraint $\mathcal{C}_i$ at level i is *constitutional* if it satisfies all three of the following:

(C1) *Intrinsic.* $\mathcal{C}_i$ is forced by the irreducible algebraic structure of S—it is not a free parameter, an external imposition, or a modelling choice.

(C2) *Independent.* $\mathcal{C}_i$ is structurally independent of constraints at other levels: it cannot be derived from, or reduced to, a combination of constraints at levels $j \neq i$.

(C3) *Binary-eliminating.* $\mathcal{C}_i$ eliminates exactly one configuration class at level i. It does not eliminate more (over-constrained) or fewer (under-constrained). How the eliminated classes are counted in the full configuration space is fixed by Postulate 2.6 below.

Remark 2.2. Condition (C1) is the most important and the hardest to fake. The test is counterfactual: *could one choose a different value without changing the algebraic structure of the system?* If yes, the constraint is not constitutional. If no—if changing the constraint requires changing what the system fundamentally is—then it is.

Remark 2.3. Condition (C2) is what distinguishes Constitutional Forcing from ordinary constraint satisfaction. A system with k coupled constraints (each derivable from a single underlying constraint) has effective depth 1, not depth k. An example is twin-prime admissibility modulo powers of 3: for $p > 3$, the pair $(p, p+2)$ avoids divisibility by 3 if and only if $p \equiv 2 \pmod 3$, and every lift of this class to modulus 9 or 27 again avoids it, so the "conditions" modulo 9 and 27 are implied by the condition modulo 3 (Section 3.2). Note also that $\theta_1 = \theta_2 = 1/2$ (checked in the accompanying Lean files), so the value $1/2$ alone cannot distinguish depth 1 from depth 2.

2.2. Constitutional Forcing.

Definition 2.4 (Constitutional Forcing)**.** A system S exhibits *Constitutional Forcing of depth k* if:

(CF1) *Configuration space.* S has a binary configuration space $\mathcal{C}_k = \{0,1\}^k$ of size 2^k at depth k. Each configuration $c = (c_1, \ldots, c_k)$ records whether the constraint at level i is satisfied ($c_i = 1$) or violated ($c_i = 0$).

(CF2) *k independent constitutional constraints.* S has exactly k constitutional constraints $\mathcal{C}_1, \ldots, \mathcal{C}_k$ in the sense of Definition 2.1, one per level.

(CF3) *Counting postulate.* Exactly k configurations of $\mathcal{C}_k$ are invalid, one attributed to each $\mathcal{C}_i$ (Postulate 2.6), so that the valid configurations number $2^k - k$ and the governing

constant is the configuration density

$$\theta_k = \frac{2^k - k}{2^k}. \tag{1}$$

(CF4) *Uniqueness.* The governing constant of S (for example, its level of distribution) equals θ_k, and it cannot be changed without either removing a constitutional constraint (reducing depth) or adding one (increasing depth).

Remark 2.5. In this revision the formula $(2^k - k)/2^k$ is part of the definition, through the counting postulate in (CF3); it is not derived. Condition (CF4) carries the substantive content: for a given system it asserts that the system's governing constant equals θ_k and is rigid. Verifying (CF4) for a particular system is a mathematical claim about that system and needs its own proof.

2.3. **The counting postulate.** The February version stated the following as a theorem (the "Counting Theorem") and gave a proof by induction. The proof is invalid (Remark 2.7), and the statement is now a postulate.

Postulate 2.6 (Counting postulate)**.** In a system of depth k, exactly k configuration classes in $\mathcal{C}_k = \{0,1\}^k$ are eliminated, one by each constitutional constraint.

Granting Postulate 2.6, the valid configurations number $2^k - k$ out of 2^k, and the configuration density is $\theta_k = (2^k - k)/2^k$, as in (1).

Remark 2.7 (Why this is not a theorem). Under (CF1), $c_i = 1$ exactly when constraint i holds. If a configuration is called invalid whenever some constraint fails, then a single failing constraint already invalidates the 2^{k-1} configurations with $c_i = 0$; in total $2^k - 1$ configurations are invalid and exactly one, $(1, \ldots, 1)$, is valid, for every k (for $k = 3$: seven invalid, not three). The revised Lean file P05 proves this count as `invalid_count`. The withdrawn induction overlooked that each of the $k - 1$ configurations invalid at depth $k - 1$ splits into two at depth k. The count k is therefore an additional assumption about which configuration classes are counted as eliminated, not a consequence of (CF1) and (CF2).

Table 1 displays the configuration densities for $k = 1, \ldots, 6$.

TABLE 1. Configuration densities $\theta_k = (2^k - k)/2^k$ under Postulate 2.6. The "Invalid" column is the postulated count k; if "invalid" meant "some constraint fails", it would be $2^k - 1$.

k	2^k	Invalid	Valid	θ_k
1	2	1	1	1/2
2	4	2	2	1/2
3	8	3	5	5/8
4	16	4	12	3/4
5	32	5	27	27/32
6	64	6	58	29/32

3. Three Candidate Instances

We examine three candidate instances against Definition 2.4. None of them currently satisfies the definition in full; for each we record what holds and what does not.

3.1. **Prime arithmetic: Bombieri–Vinogradov ($k = 1$).** The Bombieri–Vinogradov theorem [3, 12] states that for every $A > 0$ there is $B = B(A)$ such that

$$\sum_{q \le x^{1/2}(\log x)^{-B}} \max_{(a,q)=1} \left| \psi(x; q, a) - \frac{x}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A}.$$

It is a statement on average over the moduli, not one for each $q \le x^{1/2}$. The exponent $1/2$ is its level of distribution.

Proposal 3.1 (Bombieri–Vinogradov as a depth-1 system)**.** It is proposed that the Bombieri–Vinogradov level of distribution $1/2$ is an instance of Constitutional Forcing at depth $k = 1$.

Check against Definition 2.4. (CF1): Configuration space $\{0, 1\}^1 = \{0, 1\}$, size 2.

(CF2): The proposed constraint is primality. A positive integer either is prime ($c_1 = 1$) or is not ($c_1 = 0$); the constraint is intrinsic, and there is only one. But the same reasoning applies verbatim to any predicate on the integers (being squarefree, being a sum of two squares, ...), and would assign each of them depth 1 and constant $1/2$. As applied here, (CF2) therefore contains nothing specific to the primes or to the Bombieri–Vinogradov theorem.

(CF3): Under Postulate 2.6, one configuration is invalid and one valid, so $\theta_1 = 1/2$.

(CF4): Not established, and the justification given in the February version was incorrect. The Generalised Riemann Hypothesis does not raise the level beyond $1/2$: for a single modulus it gives the expected asymptotic only for $q \le x^{1/2-\varepsilon}$, and a level beyond $1/2$ for general moduli is the content of the separate Elliott–Halberstam conjecture. Levels beyond $1/2$ are known unconditionally in restricted settings: $1/2 + 1/584$ (Zhang [13]) and $1/2 + 7/300$ (Polymath [10]) for smooth squarefree moduli; and, for a fixed residue class and suitably well-factorable weights, $4/7$ (Bombieri–Friedlander–Iwaniec [4]), $3/5$ (Maynard [8]), $66/107$ (Lichtman [7]) and $5/8$ (Pascadi [9]). None of $4/7$, $3/5$ or $66/107$ has the form $(2^k - k)/2^k$, so the claim of the February version that any improvement on $1/2$ requires an increase of depth is not supported. Pascadi's theorem concerns general moduli; it does not restrict to cascade moduli and does not "increase the depth".

3.2. **Prime arithmetic: cascade moduli (proposed $k = 3$).** For cascade moduli $q = 3^K$, the companion paper [16] conjectures a level of distribution $\theta_W = 5/8$. The February version presented this as an established depth-3 instance. Both the arithmetic statement and the depth assignment are, at present, unsupported.

Proposal 3.2 (Cascade moduli as a depth-3 system)**.** It is proposed that primes have level of distribution $\theta_3 = 5/8$ to the moduli $q = 3^K$, and that this is an instance of Constitutional Forcing at depth $k = 3$. Both parts are conjectural.

Check against Definition 2.4. (CF1): Configuration space $\{0, 1\}^3$, size 8.

(CF2): The February version proposed three constraints on a twin prime pair $(p, p + 2)$ with $p > 3$, at the moduli 3, 9 and 27. Only the first is a constraint. Modulo 3, $p \equiv 2$ is the only class for which neither p nor $p + 2$ is divisible by 3. Modulo 9 and 27 nothing further is excluded: each of the 3 (respectively 9) lifts of 2 mod 3 keeps both p and $p + 2$ prime to 3. The conditions modulo 9 and 27 are implied by the condition modulo 3, so (C2) fails and, by the paper's own criterion, the depth is 1. Nor does $\mathbb{Z}[\omega]$ supply three levels: since $(3) = (1 - \omega)^2$, the $(1 - \omega)$-adic filtration of $\mathbb{Z}[\omega]/(3^K)$ has $2K$ steps, each with quotient $\mathbb{F}_3$. The "eight configuration classes" were never defined.

(CF3): With one constraint, Postulate 2.6 gives $\theta_1 = 1/2$. The value 5/8 would require three independent constraints, and none has been exhibited beyond the first.

(CF4): Not established. Twin-prime admissibility and the level of distribution of primes to the moduli 3^K are, moreover, different objects. The arithmetic statement itself is open in its meaningful form:

- The summed statement
$$\sum_{3^K \le x^\theta} \max_{(a,3)=1} \left|\psi(x; 3^K, a) - \frac{x}{\varphi(3^K)}\right| \ll \frac{x}{(\log x)^A}$$
holds for *every* fixed $\theta < 1$: for $3^K > (\log x)^{A+1}$ the trivial bound $\ll x \log x/3^K$ sums geometrically, and the Siegel–Walfisz theorem covers the $O(\log\log x)$ remaining moduli. In this form it carries no information about 5/8.
- The per-modulus version (the expected asymptotic for $\psi(x; 3^K, a)$ for each $3^K \le x^{5/8-\varepsilon}$) would go beyond what GRH gives ($x^{1/2}$). For prime-power moduli the best known ranges are 3/8 (Barban–Linnik–Tshudakov [2]), 2/5 (Gallagher [5]) and 5/12 (Huxley), with lower bounds of the correct order up to $x^{0.4736}$ (Banks–Shparlinski); see [1] for an account.

The "uniqueness of 5/8" cited in the February version (an Eisenstein "ceiling 2/3" combined with cyclic-group cancellation) is withdrawn. Pascadi's exponent 5/8 [9] concerns smooth numbers, and primes with triply-well-factorable weights, in progressions to general moduli; its numerical agreement with θ_3 is a coincidence, not evidence.

3.3. **Information theory: Shannon's theorems (a proposed analogy).** The February version presented Shannon's asymptotic equipartition property (AEP) and channel coding theorem [11] as depth-1 instances. They do not satisfy Definition 2.4, because their governing constants are not $\theta_1 = 1/2$.

- *AEP.* For a memoryless source with alphabet $\mathcal{A}$ and entropy H (logarithms to base 2), the typical set has about 2^{nH} of the $|\mathcal{A}|^n$ sequences of length n. The natural constant is the exponent ratio $H/\log|\mathcal{A}|$, which takes every value in $[0, 1]$: it is 0.469 for a Bernoulli(0.1) source and 0.811 for a Bernoulli(0.25) source. The typical set's share of $\mathcal{A}^n$ is about $2^{-n(\log|\mathcal{A}|-H)}$, which tends to 0 unless the source is uniform.
- *Channel coding.* The capacity is $C = \max_{p(x)} I(X;Y)$. For a binary symmetric channel with crossover probability ρ it is $1 - h(\rho)$, where h is the binary entropy function; again this takes every value in $[0, 1]$.

What the two theorems share with Definition 2.4 is a binary split (typical or atypical; rate achievable or not). As with primality in Section 3.1, a binary split by itself would assign depth 1 to any predicate. We therefore retain the comparison only as a proposed analogy: in both settings a condition intrinsic to the system singles out part of an exponential space, and a constant fixed by the system governs its size. Whether the analogy can be sharpened into an instance of Definition 2.4 is open; as it stands, the constants $H/\log|\mathcal{A}|$ and C do not equal θ_1.

4. Non-Instances and Falsifiability

A framework that cannot be wrong is not mathematics. For Constitutional Forcing to make predictions, the depth k of a system—its constraints, with (C2) checked—must be fixed before its governing constant is compared with θ_k. Constitutional Forcing at depth k

requires *genuinely* independent constraints, in the sense of (C2), not merely the appearance of independence; Section 3.2 gives an example where apparent independence fails. The February version did not meet this standard, and its proposed non-instance did not test the framework.

Remark 4.1 (Sophie Germain pairs). The February version treated Sophie Germain pairs $(p, 2p+1)$ as a "non-instance" of depth 2 with effective depth 1, on the ground that "$2p+1$ is odd" follows from "p is odd". That is no constraint at all: $2p+1$ is odd for every integer p. The genuine local coupling is modulo 3: for $p > 3$, the conditions $3 \nmid p$ and $3 \nmid 2p+1$ together force $p \equiv 2 \pmod 3$, the same single condition as for twin primes. Since $\theta_1 = \theta_2 = 1/2$, the example cannot discriminate between depths 1 and 2 and is not a falsification test. Moreover, a "level of distribution of Sophie Germain pairs" is not a defined notion, and no finite computation can confirm an asymptotic level, so the claimed computational confirmation is withdrawn. The accompanying Lean files check $\theta_1 = \theta_2$ and the mod-3 fact.

5. The Constitutional Forcing Conjecture

Conjecture 5.1 (Constitutional Forcing Conjecture)**.** *Let S be a system exhibiting Constitutional Forcing of depth k in the sense of Definition 2.4. Then the governing constant of S is uniquely forced to be*

$$\theta_k = \frac{2^k - k}{2^k},$$

and this value is achievable: there exist systems exhibiting Constitutional Forcing at every depth $k \geq 1$.

Remark 5.2. Conjecture 5.1 has two parts. Under Definition 2.4 the value θ_k is built into (CF3) through Postulate 2.6, so the *uniqueness* part holds by definition and carries no independent content. The substantive content is the *achievability* part: for each k, a system must be exhibited whose k constraints are fixed in advance and satisfy (C1)–(C3), and whose governing constant is proved to equal θ_k, as (CF4) requires. No depth is currently realised in this sense: the $k = 1$ candidates of Section 3 rest on binary splits that would assign $1/2$ to any predicate, and the $k = 3$ candidate is unsupported. The February version singled out $k = 4$, with governing constant $3/4$, as the first open case; in fact every depth, including $k = 1$, is open in this sense. Table 1 is the programme's prediction set, and it becomes falsifiable only when k is fixed for a system before its constant is known.

Remark 5.3. The companion paper [18] discusses a version of this conjecture in the terminology of the Shannon–Wakil Effect. The February version of this remark attributed to [18] a category of Shannon–Wakil systems and a functor to $(0, 1)$; no such construction appears there, and none is claimed here.

6. Position in the Nine-Paper Programme

This paper is p01, the first of a nine-paper programme. Table 2 lists the papers with the status of their main claims after the November 2026 revision.
The logical dependency structure is:

$$\text{p01} \longrightarrow \begin{cases} \text{p02, p05, p06, p08, p09} & \text{(proposed instances and comparisons)} \\ \text{p03} & \text{(methodology)} \\ \text{p04, p07} & \text{(arithmetic of the moduli } 3^K\text{)} \end{cases}$$

TABLE 2. The nine-paper programme and the revised status of each paper's main claim.

Paper	Title	Main claim and revised status
p01	Constitutional Forcing (this paper)	Framework; the count k is a postulate; no instance established
p02	Khayyam's Triangle: Historical Restoration and the Geometry Hidden in the Numbers [14]	History of the arithmetic triangle and its mod-3 fractal; the claimed link between the fractal and θ_k is withdrawn
p03	Spherical Cow Philosophy: A Methodological Framework for Mathematical Discovery [15]	Methodology; its arithmetic examples are corrected
p04	Ternary Forcing on the Hexagonal Lattice: Eisenstein Integers, Cascade Moduli, and a Conjectured Level of Distribution 5/8 [16]	$\theta_W = 5/8$ for the moduli 3^K: a conjecture
p05	Ramanujan's Dimensional Forcing: A Universal Formula for Sieve Levels [17]	The formula $\theta_k = (2^k - k)/2^k$, as a postulate
p06	The Shannon–Wakil Effect: Constitutional Forcing as a Universal Pattern in Information Theory and Prime Arithmetic [18]	Proposed analogy between Shannon's theorems and the θ_k scheme
p07	Condition W3: Absence of Siegel Zeros for Characters Modulo 3^K [19]	True, with a one-line proof from the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^\times$; affects no level of distribution
p08	The Lemke Oliver–Soundararajan Bias and Ternary Constitutional Structure [20]	The bias in consecutive primes; the claimed ternary explanation of it is withdrawn
p09	The Constitutional Constant $\theta_{\mathrm{FFT}} = 1/2$: Conjugate Symmetry of the Real Discrete Fourier Transform [21]	The DFT of a real signal of length N has N independent real parameters (a standard fact); its depth is not identified, since $\theta_1 = \theta_2$

A reader interested in the arithmetic should read p01, p05, p04 and p07 in order. A reader interested in the cross-domain comparisons should read p01, p02, p03, p06 and p09.

7. What Constitutional Forcing Is Not

7.1. **Not, at present, more than an analogy across domains.** The February version claimed that Constitutional Forcing is not an analogy between number theory and information theory but a common, causal mechanism that both instantiate. That claim is withdrawn. Section 3.3 shows that the information-theoretic constants are not θ_1; the comparison is a proposed analogy, and no causal link is claimed.

7.2. **Numerology, and how to avoid it.** The formula $\theta_k = (2^k - k)/2^k$ is not derived from the definitions: it is the counting postulate (Postulate 2.6), and the February claim that "the formula is a theorem, not a fit" is withdrawn. Whether the framework is more than numerology therefore depends on whether k can be fixed for a system independently of the

constant to be explained. The risk is concrete: if k or the size of the configuration space may be chosen after the target value is known, the formula can be made to match many constants, and since $\theta_1 = \theta_2 = 1/2$ the value $1/2$ cannot identify the depth at all. A candidate instance should therefore (i) use the configuration space $\{0,1\}^k$; (ii) fix k and the constraints, with (C2) checked, before any comparison with a known constant; and (iii) prove (CF4) for the system. None of the candidates in Section 3 meets all three conditions at present, and the Sophie Germain example (Remark 4.1) is not a test.

7.3. **Not a claim of isomorphism.** We do not claim that prime arithmetic and information theory are isomorphic, or that there is a functor between them as mathematical objects. The February version claimed that both exhibit Constitutional Forcing; in this revision we claim only a proposed analogy (Section 3.3). Two physical systems that both obey Hamiltonian dynamics share a structural description without being isomorphic; here, even the shared description is not yet established.

Acknowledgements

The programme owes its inspiration to Omar Khayyam (c. 1048–1131), after whom the arithmetic triangle is named in Iran, and to Claude Shannon, whose 1948 theorems prompted the cross-domain comparison of Section 3.3.

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The ARC Institute of Knowware, Calgary, Alberta, Canada
Email address: the@knowware.institute